

PYTHON FOR BEGINNERS · DAY 3

if · elif · else

Teaching Python to make decisions

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```
age = 20
if age >= 18:
    print("Adult")
else:
    print("Under 18")
```

BY THE END OF TODAY

You will be able to…

- Make Python make decisions
- Compare values
- Build interactive programs
- Use **if**, **elif** and **else**
- Use **and**, **or**, **not**
- Create a simple game 🎮

PYTHON SPEED ROUND · NO LAPTOPS!

5 seconds each — hands up! 🙋

Q1. Which is Python?

A: print("Hello")
B: say("Hello")
C: display("Hello")
D: show("Hello")

Q2. Find the error

```
name = "Homaira"
Print(name)
```

Q3. Find the error

```
age = input("Age: ")
print(age + 5)
```

BUT THERE IS ONE PROBLEM…

Real programs make decisions



Weather

If it’s raining → take an umbrella



Grades

If score ≥ 50 → print “Passed”



Access

If age ≥ 18 → allow registration

So far: **input** → **calculate** → **print**. Today we teach Python to **think**! 🧠

CONDITIONS

A yes/no question: True or False

Operator	Meaning	Example
==	equal to	5 == 5 → True
!=	not equal to	5 != 3 → True
>	greater than	7 > 10 → False
<	less than	3 < 8 → True
>=	greater or equal	18 >= 18 → True
<=	less or equal	20 <= 15 → False

⚠️ = vs ==

`age = 18`

one = **stores** a value

`age == 18`

two == **compares** values

THE IF STATEMENT

Run this only if it’s True

```
age = 20
if age >= 18:
    print("You are an adult.")
```

```
You are an adult.
```

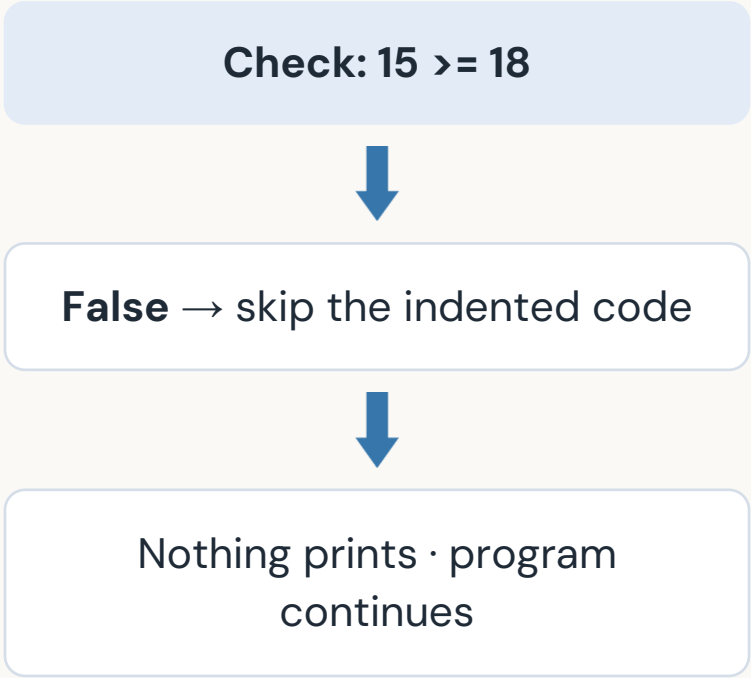
1. Store 20 in age
2. Check: 20 >= 18 → True
3. Run the indented line

Every **if** needs a colon **:** and the code inside is indented **4 spaces**.

WHEN THE CONDITION IS FALSE

Python skips the block

```
age = 15
if age >= 18:
    print("You are an adult.")
```



🧠 PREDICTION CHALLENGE

What will this code print?

```
temperature = 35
if temperature > 30:
    print("It is hot!")
print("Have a nice day!")
```

A · Nothing

B · “It is hot!”

C · “It is hot!” and “Have a nice day!”

D · “Have a nice day!”

IF + USER INPUT

Did the student pass?

```
score = int(input("Enter your score: "))
if score >= 50:
    print("You passed!")
```

Input	Condition	Output
75	True	You passed!
40	False	(nothing)
50	True	You passed!

Use **int()** — input() gives text, and you can’t compare text with a number.

🐛 COMMON MISTAKES

Spot the bug

Forgot int()

```
age = input("Age: ")
if age >= 18:
```

Forgot the colon

```
if age >= 18
    print("Adult")
```

Forgot to indent

```
if age >= 18:
print("Adult")
```

= instead of ==

```
if age = 18:
    print("Exactly 18")
```

ELSE — THE OTHER CHOICE

If not, do this instead

Code

```
age = 15
if age >= 18:
    print("You are an adult.")
else:
    print("You are under 18.")
```

Output

```
You are under 18.
```

else has **no condition**. It runs when the if is False.

ELIF — MULTIPLE CHOICES

More than two outcomes

```
score = 75
if score >= 80:
    print("Grade A")
elif score >= 70:
    print("Grade B")
else:
    print("Keep practicing!")
```

1. 75 >= 80 → False · skip

2. 75 >= 70 → True · run it!

3. Prints **Grade B** and stops checking

elif = “else if”. You can have as many as you need.